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central nervous system (CNS) through a motor neuron. The junction between the motor neuron and the sarcolemma of the muscle fibre is called the neuromuscular junction or motor end plate. When nerve impulse reaches a neuromuscular junction, acetylcholine is released. It initiates the opening of multiple gated channels in sarcolemma. The action potential travels along the T-tubules and triggers the release of calcium ions from the sarcoplasmic reticulum. The released calcium ions bind to troponin on thin filaments. The tropomyosin uncovers the myosin-binding sites on thin filaments. Now the active sites are exposed to the heads of myosin to form a cross-bridge. During cross-bridge formation actin and myosin form a protein complex called actomyosin. Utilizing the energy released from hydrolysis of ATP, the myosin head rotates until it forms a 90° angle with the long axis of the filament. In this position myosin binds to an actin and activates a contraction – relaxation cycle which is followed by a power stroke.

The power stroke (cross-bridge tilting) begins after the myosin head and hinge region tilt from a 90° angle to a 45° angle. The cross-bridge transforms into strong, high-force bond which allows the myosin head to swivel. When the myosin head swivels it pulls the attached actin filament towards the centre of the A-band. The myosin returns back to its relaxed state and releases ADP and phosphate ion. A new ATP molecule then binds to the head of the myosin and the cross-bridge is broken. At the end of each power stroke, each myosin head detaches from actin, then swivels back and binds to a new actin molecule to start another contraction cycle. This movement is similar to the motion of an oar on a boat. At the end of each power stroke, each myosin head detaches from actin, then swivels back and binds to a new actin molecule to start another contraction cycle. The power stroke repeats many times until a muscle fibre contracts. The myosin heads bind, push and release actin molecules over and over as the thin filaments move toward the centre of the sarcomere. The repeatedly formation of

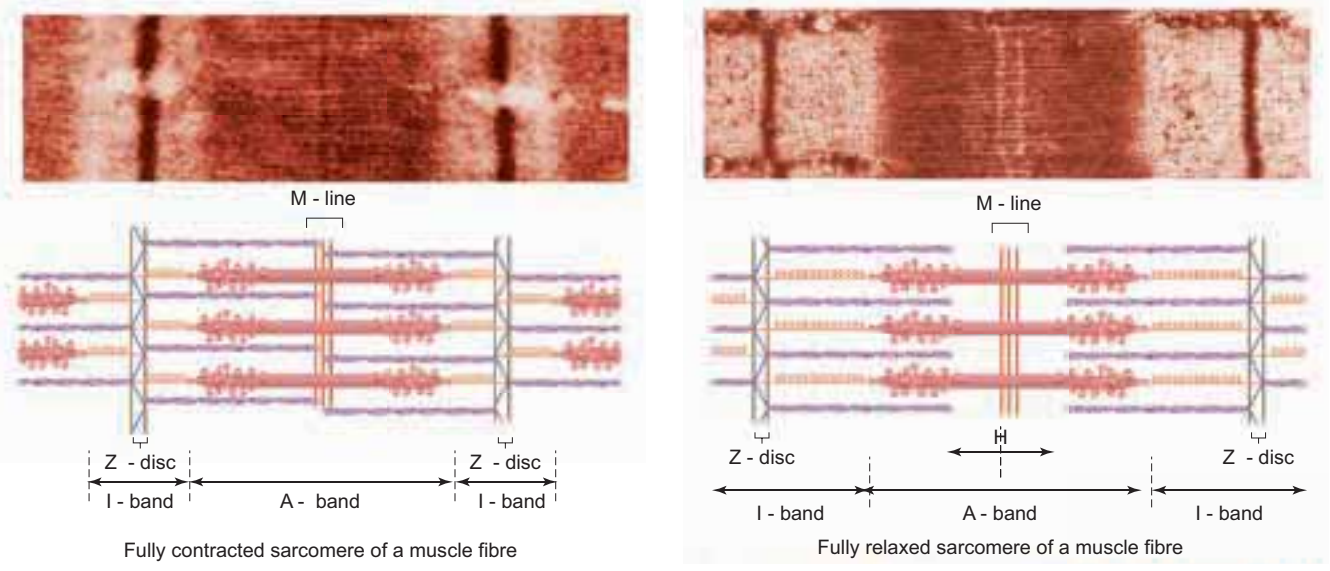


Figure 9.5 Sliding filament model of muscle contraction